

Java Programming
Exam #1

Name: George Westfall (.5 point)
Date: 10/11/25 (.5 point)

1. For each of the following variable names, circle "legal" or "error". (12 points)

a.) firstName legal error

d.) \$uffix legal error

b.) la\$tName legal error

e.) &SSN legal error

c.) 9LIVES legal error

f.) _Emp_Num legal error

2. Choose meaningful names and write a single variable declaration for each of the following values. (12 points)

a.) Net income on paycheck.

`double netIncome;`

b.) Floor number in a building.

`int floorNumber;`

c.) Fire alarm is on or off

`boolean alarmSet = true;`

d.) Letter on a computer keyboard.

`char letter;`

3. Rewrite the Java program with corrected syntax using the space provided below. (15 points)

```

1  import java.util.Scanner;
2
3  //
4  * Jim Smith
5  * File: DaysToSeconds.java
6  //
7  public class DaysToSeconds {
8      public static void main(String args) {
9          Scanner in = new Scanner(System.in);
10         final int HOURS_IN_DAY = 24;
11         final int MINUTES_IN_HOUR = 60;
12         final int SECONDS_IN_MINUTE = 60;
13         int secondsInDay = HOURS_IN_DAY * MINUTES_IN_HOUR * SECONDS_IN_MINUTE;
14         double totalDays 0.0;
15         double totalSeconds 0.0;
16
17         System.out.print('Enter the number of days to convert to seconds: ')
18         totalDays = in.nextDouble();
19
20         totalSeconds = totalDays * secondsInDay;
21
22         System.out.print(totalDays " days is equal to ");
23         System.out.println(totalSeconds + " seconds.");
24     }
25 }

```

```

1  /x
2  * JIM SMITH
3  * FILE: DAYSTOSECONDS.java
4  /
5
6  import java.util.Scanner;
7
8  public class DaysToSeconds {
9      public static void main(String args) {
10         Scanner in = new Scanner(System.in);
11         final int HOURS_IN_DAY = 24;
12         final int MINUTES_IN_HOUR = 60;
13         final int SECONDS_IN_MINUTE = 60;
14         int secondsInDay = HOURS_IN_DAY * MINUTES_IN_HOUR *
15             SECONDS_IN_MINUTE;
16         double totalDays = 0.0;
17         double totalSeconds = 0.0;
18
19         System.out.print("Enter the number of days to convert to
20             seconds: ");
21         totalDays = in.nextDouble();
22         totalSeconds = totalDays * secondsInDay;
23         System.out.print(totalDays + " days is equal to ");
24         System.out.println(totalSeconds + " seconds.");
25     }

```

4. Do a hand simulation. (10 points)

```
public class ExamVariables {  
    public static void main(String args[]) {  
        int gamma = 6;  
        int delta = 12;  
        int zeta = 0x40;  
        char alpha = 'z';  
        char beta = '2';  
        char phi = 'B';  
  
        System.out.println("1. " + (zeta / delta) + " " + (delta / gamma));  
        System.out.println("2. " + zeta + " " + (delta + zeta));  
        System.out.println("3. " + (beta + 5) + " " + (beta + 14));  
        System.out.println("4. " + (char)(alpha - beta) + " " + (phi - '!'));  
    }  
}
```

Memory

gamma	6
delta	12
zeta	64
alpha	122
beta	50
phi	66

Screen

```
1. 5 2  
2. 64 76  
3. 55 64  
4. H 35
```


5. The equation to calculate the area of a trapezoid is $a = \left(\frac{b_1 + b_2}{2}\right) * h$, where a is the area of the trapezoid, b_1 is the length of the first base, b_2 is the length of the second base, and, h is the height. Write an interactive Java class named TrapezoidArea which prompts the user to enter a value for the integer variables base1, base2, and height. The program will then calculate the area of the trapezoid as a double and print the result to the screen. (For the screen simulation, have the user type in 14 inches for base1, 10 inches for base2, and 5 inches for the height) (20 points)

```
import java.util. Scanner ;

/*
 * Calculate area of a trapezoid; a = ((b1 + b2) / 2) * h
 */
public class TrapezoidArea {
    public static void main(String args[]) {
        Scanner in = new Scanner(System.in);

        int base1;
        int base2;
        int height;
        System.out.print("Enter the length of the first base: ");
        base1 = in.nextInt();
        System.out.print("Enter the length of the second base: ");
        base2 = in.nextInt();
        System.out.print("Enter the height in inches: ");
        double area = ((double)(base1 + base2)/2) * height;
        System.out.println("\n --- Calculation Result ---");
        System.out.printf("Area = %0.2f square inches\n", area);
        height = in.nextInt();
    }
}
```

Screen

```
Enter the length of the first base: 14
Enter the length of the second base: 10
Enter the height in inches: 5

--- Calculation Result ---
Area = 60 square inches
```

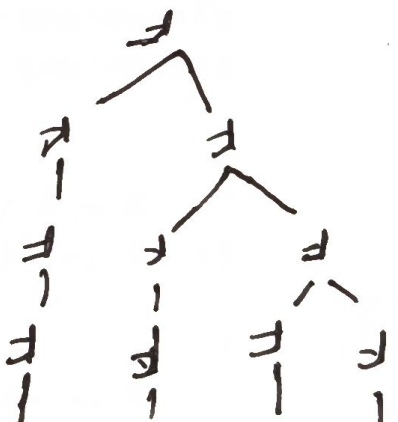
6. Given the following values for boolean variables p, q, r, s, evaluate each condition. (16 points)

f = false, g = false, h = true, i = true

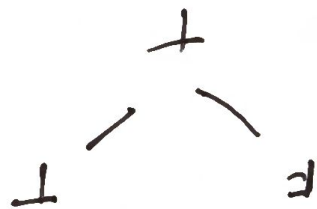
a.) $f \vee g \wedge h \vee i$



b.) $f \wedge g \wedge h \wedge i$



c.) $(f \wedge g) \vee (h \wedge i)$



d.) $i \vee (f \vee g) \wedge h \wedge i$



7. Do a hand simulation of this program. (15 points)

```
public class Exam1Quest7 {  
    public static void main(String args[]) {  
        int x = 7;  
        int y = 1;  
  
        if (x != y) {  
            x *= y++;  
        }  
  
        if (y <= x) {  
            y *= x;  
        } else {  
            x *= y;  
        }  
  
        if (x != y) {  
            y = y / (y / x);  
        }  
  
        System.out.println("x: " + x);  
        System.out.println("y: " + y);  
    }  
}
```

Memory

x 7
y 1 2, 14, 7

Screen

x: 7
y: 7

BONUS (5 points)

Write the Java class HelloWorld that outputs the string "Hello World!" to the screen.

```
public class HelloWorld {  
    public static void main (String args[]) {  
        System.out.println("Hello World");  
    }  
}
```